Project 1 data card

properties_modeling_v4.csv

Dataset: properties_modeling_v4.csv

Student release of the Model 1 table (v3 build). One file; splits are pre-assigned (incident-grouped).

Splits (do not reshuffle)

split	Role
`train`	Fit models, compute imputation/encoding statistics
`selection`	Compare models, thresholds, hyperparameters, feature choices
`assessment`	**One final report** — touch only after you lock your model

Outcome columns

Column	Description
`damage_category`	DINS damage label (5 levels; use for baseline rates)
`damage_ordinal`	0 = No Damage … 4 = Destroyed
`damage_ratio_mid`	Midpoint damage fraction proxy
`serious_damage`	Binary (Major/Destroyed = 1) — common classification target

Feature blocks (map to CDI modules)

- **Structure s:** `structure_*`, `roof_construction`, …, `propane_tank_distance`, outbuildings, `cars_on_property`
- **Exposure x:** `latitude`, `longitude`, `county`, `city`, `community`, `assessed_improved_value`, `elevation_m`, `slope_deg`, …
- **Static hazard H:** `fhsz_category`, `wrc_burn_probability`, `wrc_risk_to_potential_structures`
- **Event fire F:** `fire_inside_perimeter`, `fire_distance_km`, `fire_perimeter_acres`

Original data sources (for feature documentation)

Layer	Source
DINS inspections	CAL FIRE Damage Inspection (DINS)
Fire perimeters (event F)	California Historical Fire Perimeters
FHSZ	CAL FIRE Fire Hazard Severity Zones
WRC hazard rasters	USFS Wildfire Risk to Communities (RDS-2020-0016-2)
Elevation / slope	USGS 3DEP Elevation

Columns to treat carefully

- **Identifiers (not features by default):** `property_id`, `global_id`, `incident_id`, `incident_name`, address fields, `apn`
- **Post-event (justify if used):** `defensive_actions`, `where_fire_started_on_structure`, `what_fire_started_from`
- **Metadata:** `fire_match_method`, `split`

Caveats

- Conditional sample: structures inspected in/near wildfires.
- Many structure fields are inspector-recorded after the fire.
- ~5% of rows lack a matched fire perimeter (`fire_distance_km` missing).